reliability. Continuous innovation in battery technology, lightweight materials, and AI optimization is essential to overcome these hurdles.

6.6.2 Operational limitation

Despite enhanced automation, fifth-generation drones still face operational constraints, particularly in terms of range, payload capacity, and endurance. Weather conditions such as heavy rain, fog, and high winds limit flight capabilities and sensor performance. Swarm coordination in real-world environments can also lead to interference, collision risks, and signal synchronization issues. Moreover, dependence on stable communication networks, such as 5G or satellite links, restricts deployment in remote or underdeveloped regions.

6.6.3 Cybersecurity and electronic warfare vulnerabilities

With the increased use of AI, cloud connectivity, and networked swarm intelligence, cybersecurity threats have become a major concern. Fifth-generation drones are susceptible to hacking, data interception, GPS spoofing, and electronic jamming. Adversaries can exploit vulnerabilities in communication links or onboard software to gain unauthorized control or disrupt missions. In military applications, such vulnerabilities pose severe risks to operational security. Therefore, incorporating secure encryption, blockchain-based authentication, and adaptive countermeasure systems is crucial for ensuring resilience.

6.6.4 Cost and accessibility issues

The development and deployment of fifth-generation drones involve high costs due to advanced materials, sensors, AI processors, and specialized manufacturing requirements. This limits their accessibility for small enterprises, academic research, and developing nations. Maintenance,